

Introduction to Razor Pages, Advantages of Razor Pages, Razor Syntax, Page model, Property binding, Types of Views, Partial Views, Routing,

MVC, Model binding, TagHelpers, HTML Helper classes, View Categorization based on Model Validations Overview , Validations with Data Annotation , Server Side and Client Side Validation , Custom Server side validation , Model level validation using IValidatableObject , Custom unobstrive Client side Validation, Remote Validation, Url Routing Overview, Custom Routes, Attribute Routing , Routing Constraints

What are the different types of [ASP.NET](#) Web Application templates ?

<u>MVC Web App (Fullstack)</u>	<u>Web API(backEnd)</u>	<u>Web Application(Razor pages -responsive UI)</u>
- Combines UI + Business Logic + Data Handling - Uses Model-View-Controller pattern - HTML Views - Enterprise apps	- Used for building RESTful services - No UI, - returns JSON/XML - Microservices, APIs	- Page-based approach - Simplified UI + backend integration - HTML Pages - Lightweight apps

What are RAZOR pages ?

Razor page is a page based programming model in [ASP.NET](#) Core.
Here Each page has their own logic and UI combined (C# + HTML).CSHTML

Features

- Cleaner structure (page-focused)
- Less code compared to MVC
- Built-in model binding
- Easy to maintain

Project Structure

- Pages/Index.cshtml (UI)
- Pages/Index.cshtml.cs (Logic)

What are different stages of a page model in Razor pages ?

Lifecycle of Razor Page Model

Definition

Page Model lifecycle defines how a request is processed.

Stages

1. OnGet() - Handles GET request
2. OnPost() - Handles POST request
3. OnPut() / OnDelete() - Optional handlers
4. Model Binding - Data mapping from UI
5. Validation Execution
6. Response Rendering

Application Scenario

- User searches student via **GET (query string)**
- User submits student form via **POST**

Request /response Flow:

**Request → Routing → Handler Method → Model Binding
→ Validation → Response**

What is a property Binding ? How can we use it ?

- GET Binding : Gets data from URL
- POST binding : Gets data from form

Note: [Bind Property] : it is used for automatic mapping

GET → Search / Filter

POST → Submit / Save

One way binding	Two Way binding
Data flows in one direction (UI → Backend OR Backend → UI)	Data flows in both directions (UI ↔ Backend)
Single direction	Bi-directional
Display data OR send data	Form input + display updated value
Showing value: @Model.Name	Input + update: <input asp-for="Name" />
Display user name	Edit + submit + show updated name

Benefits of routing in MVC razor application ?

Different types of layered architecture MVC, MVVM, MVT(Python) Why do we need them ?

ex APSTNDP → PDNTSPA

What is the purpose of validation in a web application?

Different types of validation in MVC applications ?

Getting started with MVC Application in [ASP.NET](#) :

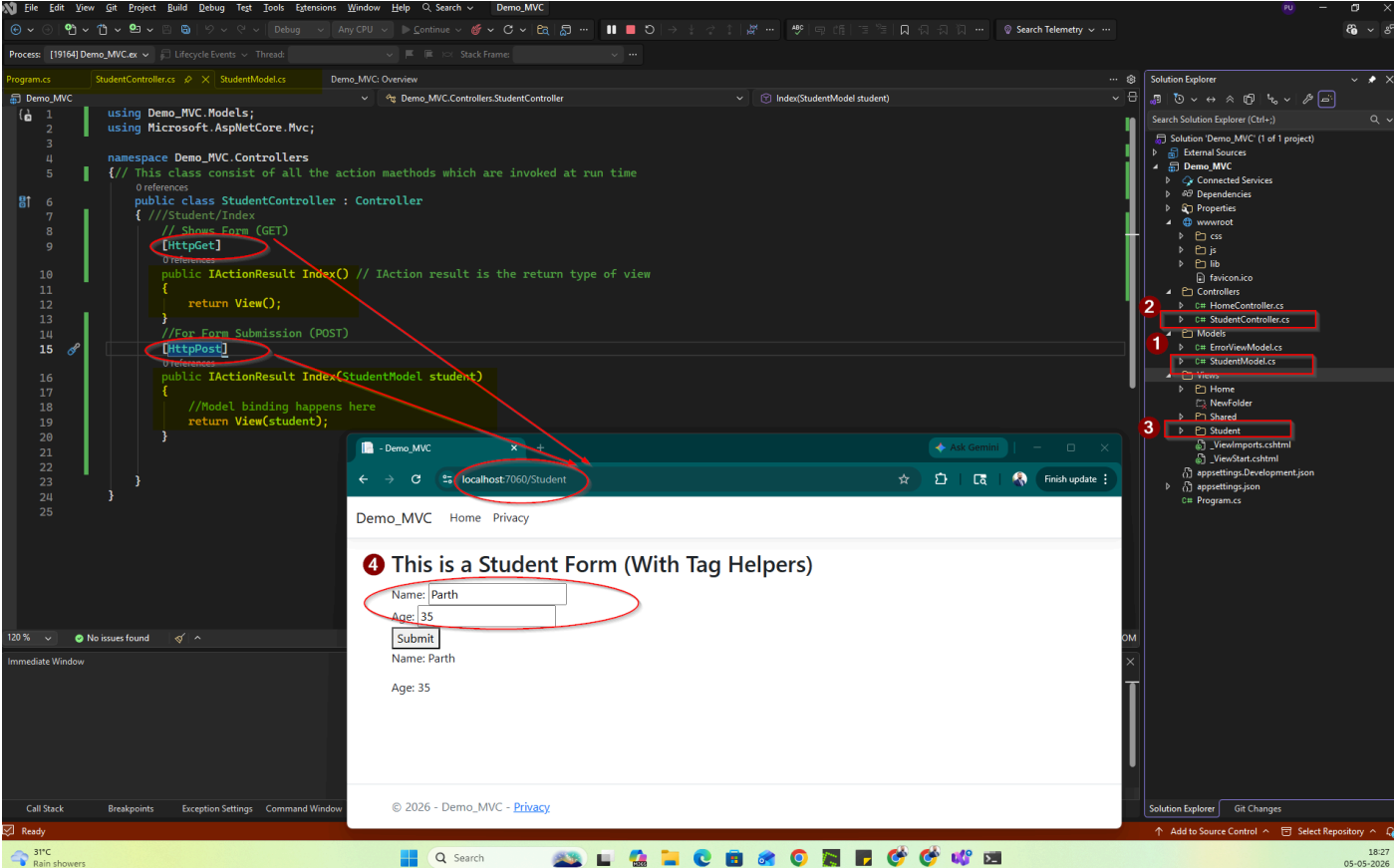
Step 1: Create a Model (Emp/[Student.cs](#))

Step 2: Create a Controller ([StudentController.cs](#))

Step 3: Create Views(Index.cshtml)

Step 4: Using html helper classes

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The screenshot displays the Visual Studio IDE with the following components:

- Code Editor:** Shows the `StudentController.cs` file. The `Index` action is defined with two methods:


```

      [HttpGet]
      public IActionResult Index() // IActionResult is the return type of view
      {
          return View();
      }

      [HttpPost]
      public IActionResult Index(StudentModel student)
      {
          //Model binding happens here
          return View(student);
      }
      
```
- Solution Explorer:** Shows the project structure. The `StudentController.cs` file is highlighted with a red circle and the number 2. The `StudentModel.cs` file is highlighted with a red circle and the number 1. The `Student` folder in the `Views` directory is highlighted with a red circle and the number 3.
- Browser Preview:** Shows the rendered form at `localhost:7060/Student`. The form title is "4 This is a Student Form (With Tag Helpers)". The form contains:
 - Name:
 - Age:
 - Submit button
 - Below the button, the form displays: Name: Parth, Age: 35

Model-View-Controller

